

First Steps in Constructing an AI-Powered Digital Twin Teacher: Harnessing Large Language Models in a Metaverse Classroom

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ABSTRACT

This study proposes a ground-breaking idea at the intersection of Artificial Intelligence and virtual education: the creation of an AI-powered Digital Twin instructor in a Metaverse-based classroom using Large Language Models. We aim to build a teacher avatar capable of dynamic interactions with students, tailored teaching approaches, and contextual response inside a virtual world. The research aims to address two major issues for both students and teachers: the Digital Twin can provide feedbacks to resolve doubts about course content and material; also, it can improve student management and allow teachers to answer the trickiest questions raised by students.

Index Terms: Social and professional topics—Professional topics—Computer education programs—Software engineering education; Computing methodologies—Artificial intelligence—Natural language processing—Discourse, dialogue and pragmatics Applied computing—Education—Computer-assisted instruction

1 INTRODUCTION

Generative AI is a broad field of artificial intelligence that includes a variety of strategies and models for generating data or content [4]. These models may generate new content from scratch, typically without direct human input or in response to precise requests. Large Language Models (LLMs) are a subset of Generative AI: they are Neural Networks trained on massive volumes of textual data collected from many sources, including online blogs, code repositories, open data, and media content. Their training is designed to teach them how to understand and generate human language. These models have shown amazing abilities in a variety of natural language processing tasks, including text production, translation, summarization, and others.

Between the different application of LLMs, we focus on Education and Training tasks, in particular in the Software Engineering topic. The main goal is to develop a digital twin of a professor, able to answer simple questions made by the course students, in an environment built in the metaverse. In this way, students can obtain a first feedback to solve their doubts regarding the course material. On the other hand, professors can better manage their time and workflow, receiving and answering only the more difficult questions.

As far as we know, this is a completely new approach in the Software Engineering Education topic. Various attempts have been conducted to generate the digital twin of a teacher: authors of paper [5] propose the digital twin of an educational institution to

overcome the constraints made by COVID-19 pandemic. A similar analysis is conducted in [8]: a virtual classroom is implemented to monitor and improve public school courses. The most similar approach to the one proposed in this paper is contained in iSTAR framework [3]: this framework aims to extend the human-machine collaboration and synergy in education and collaborative teaching, exploring the advantages of integrating machines into a teaching course.

Our paper is different from the ones explained above because it takes into account the possibility to adopt LLMs to generate a more complete Digital Twin of a university teacher. Students can also interact with this teacher using immersive technologies such as Virtual Reality (VR). The final goal is to let students explore their university in the Metaverse and chat or talk with a teacher to solve their doubts about the course material.

2 OUR APPROACH

We divide our research in three parts: identification and training of a Large Language Model, development of a Metaverse platform, and connection of the Metaverse-based avatar to the LLM through APIs. Fig. 1 shows the details of each step.

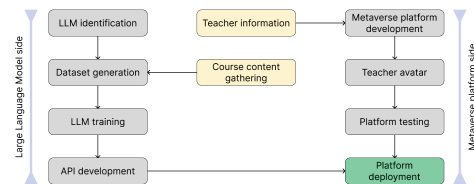


Figure 1: Overview of the research steps

2.1 LLM identification and training

The first step to follow to generate an AI-powered Digital Twin of a teacher is to find the most suitable LLM between the ones available online. There are various LLMs, each with its own architecture and parameters. Some notable ones include Generative Pre-trained Transformer (GPT), Bidirectional Encoder Representations from Transformers (BERT) and Llama. GPT is a model capable of generating human-like text without the need of a specific context. The last iteration of the model, GPT-4, can generate highly coherent and contextually relevant text, together with multimedia analysis [6]. BERT, on the other side, considers both preceding and following words to improve the output. It is lightweight and easy to use for classification tasks [2]. Llama is defined as a collection of pretrained and fine-tuned LLMs, with a scale from 7 billion to 70 billion parameters. It has been developed by Meta in 2023. Table 1 shows some models that fit our task, together with considerations on their parameters, the price required to use the model, the possibility to

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Table 1: Comparison of LLMs available online

Model name	Parameters	Price	API	Offline
GPT-3	Not known	Free as chatbot	Yes, \$	No
GPT-4	Not known	\$20/month	Yes, \$	No
LLaMa 2	7b - 13b - 70b	Free	Yes	Yes
BERT	110M - 340M	Free	Yes	Yes
Falcon	Up to 180b	Free	Yes	Yes
Mistral	7b	Free	Yes	Yes

connect through APIs and the possibility to install the model in a local environment.

We choose Mistral 7b as reference LLM because it is lightweight and it becomes easier to train it based on our needs. Anyway, we plan to test our approach with different LLMs.

The training phase is made by different tasks: the dataset is generated from the course material written for a Software Engineering course in the bachelor degree of Computer Science. The material is given as a single PDF file, then it is split into chunks of a defined number of tokens, where a token is equal to 0.75 words. These chunks are useful for the information retrieval part; a single chunk should be relevant to different user queries. Each chunk is then translated into an embedding, then all the embeddings are inserted into a vector database. This database can be queried using students prompts, then these prompts are embedded and checked for the most similar embedding present into the vector database. The database output can be given to the LLM as context for formulating the response. Fig. 2 shows the steps explained above.

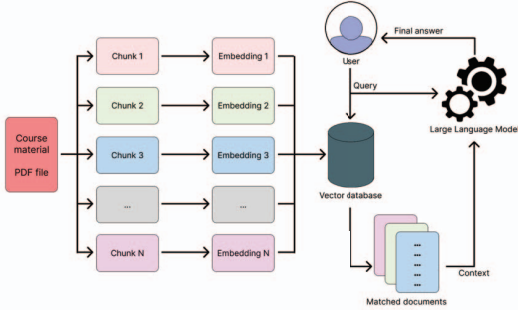


Figure 2: Details of the LLM training phase

The chosen LLM is used locally thanks to a GUI available on GitHub¹, that also makes it possible to create a local server and send prompts to the LLM using some predefined APIs.

2.2 Metaverse implementation

Metaverse is particularly useful in immersive and interactive contexts, such as tourism [1] or education [7]. In the metaverse, the student will be able to formulate a question to the avatar of the teacher and receive an answer, generated by the LLM system. During the session, only the student is connected from his/her device. The question can be either written through a keyboard or expressed verbally. In fact, the metaverse platform could be exploited not only by a Head Mounted Display (HMD), but also by a desktop or mobile app because, currently, most of the students do not own an HMD. The metaverse platform will also give the possibility to have more than one student interacting with the teacher in the same room, stimulating a collaborative approach to the learning. Furthermore, it will be possible to

interact also with the avatar of a fictitious student who suggests some frequently asked questions. The metaverse platform will be developed on Spatial and the integration with the LLM system will be managed in Unity through the Spatial Creator Toolkit. A storyboard depicting a prototype of the platform is shown in Fig. 3.



Figure 3: Prototype of the metaverse platform

The input to the LLM is sent through API calls in the form:

User input: {*USER-INPUT*}
Context: {*CHUNK-RETRIEVED-FROM-VECTOR-DB*}

3 CONCLUSION

The proposed approach aims to build the Digital Twin of a teacher, capable of giving an initial feedback to students doubts thanks to its underlying layer made by a LLM, in an immersive way thanks to the use of the Metaverse. Education can be improved with these technologies, both for teachers and students. Teachers can filter requests and improve their time management; students do not need to wait for an answer by the teacher, and obtain a first feedback from the Digital Twin and solve their doubts in a faster way.

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¹<https://github.com/oobabooga/text-generation-webui>